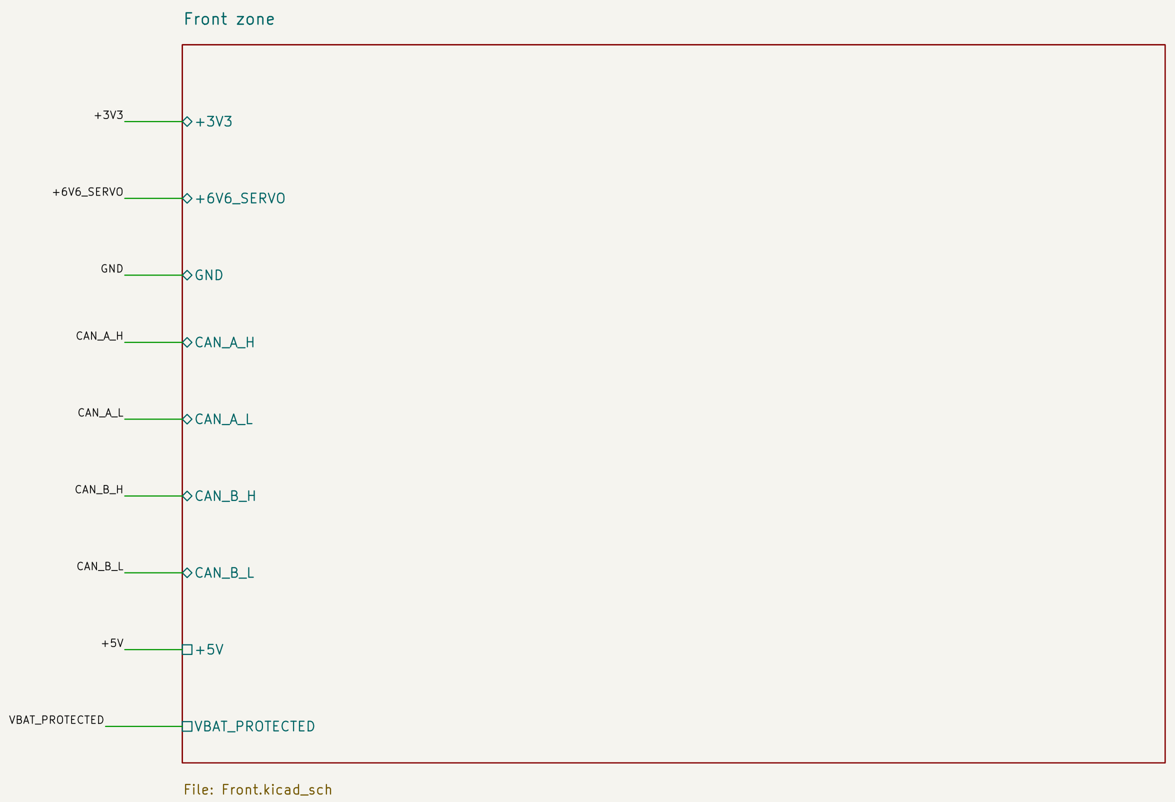
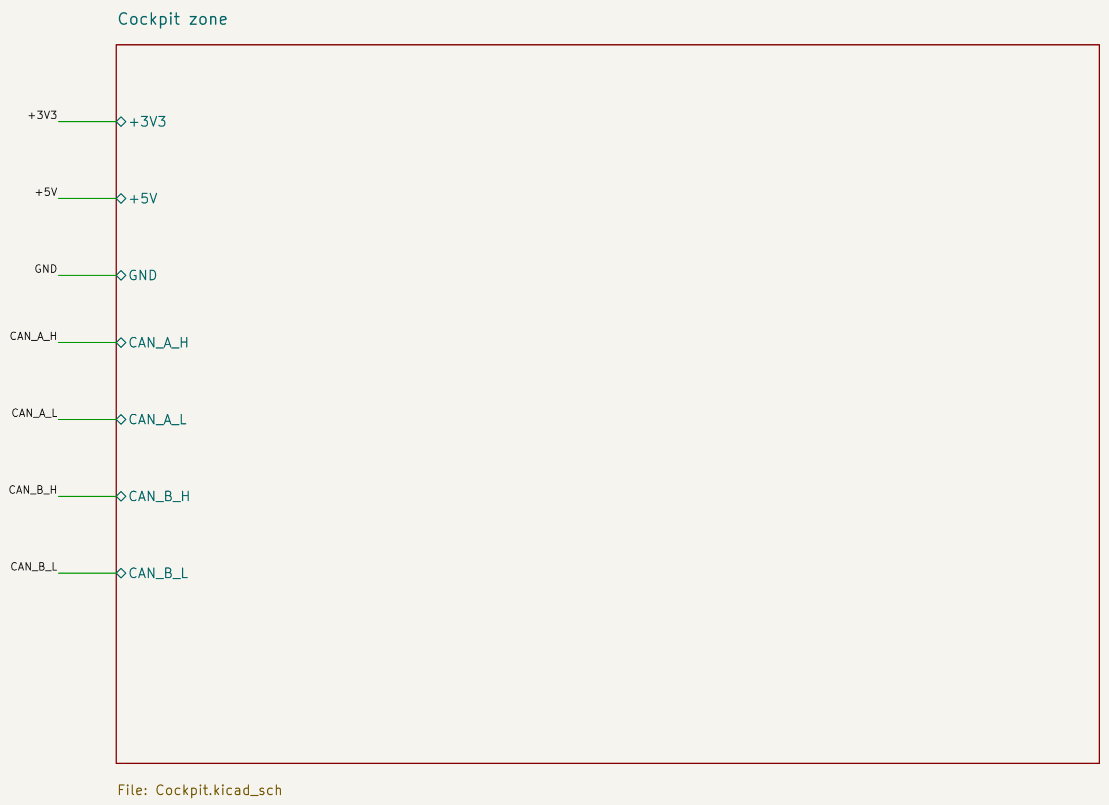
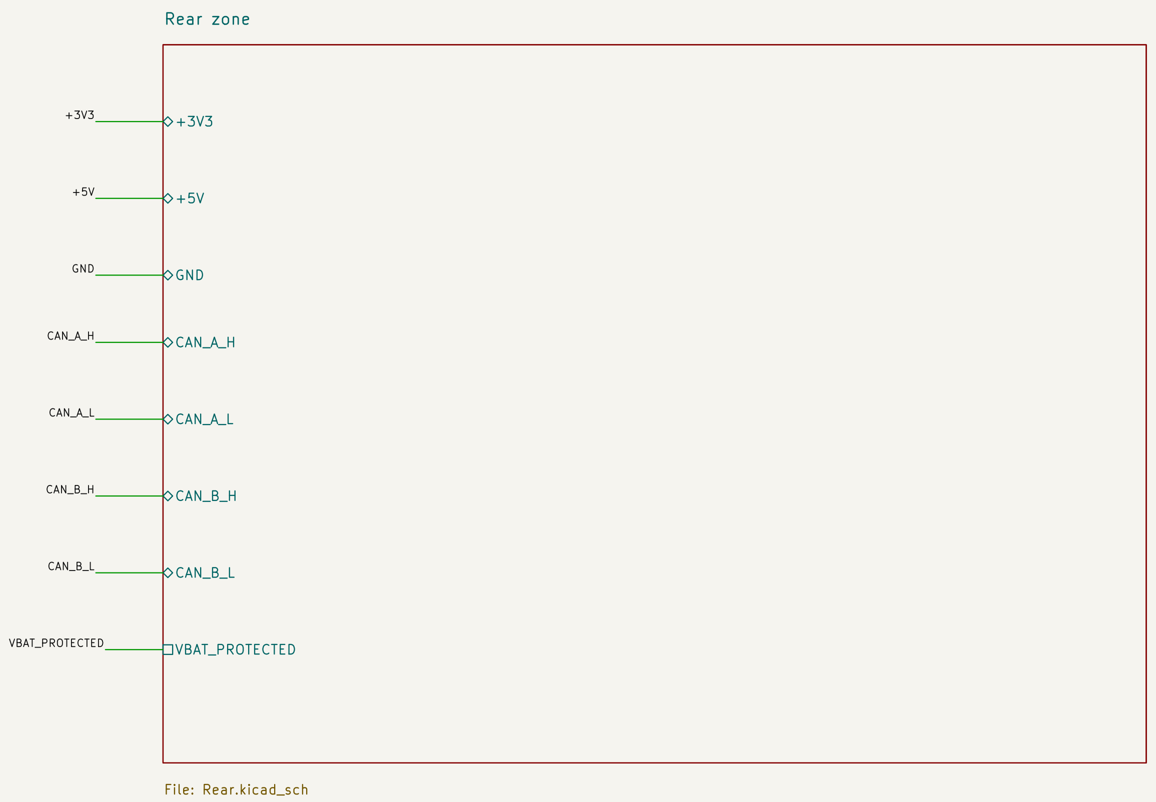
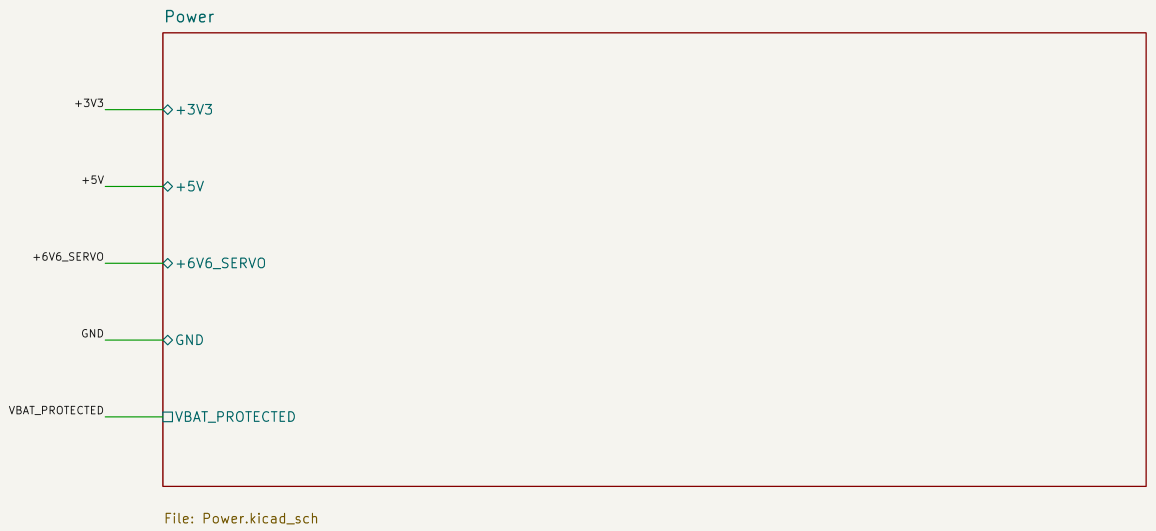
LAB 4 – CONNECTIVITY OVERVIEW

Front and rear are the terminated ends. Cockpit taps both backbones.
Each zone has its own two isolation groups; termination remains on the bus side.
Actuator connectors / motor protection / physical layout remain unfinished.









CAN_A_H/L and CAN_B_H/L are separate root nets. Same–name local labels on this root join the shown sheet pins. Child labels connect only through those ports.

Power flags document external battery power and outputs after passive filters; they do not prove current capacity, regulation, thermal behavior or PCB completeness.

See Verification.txt and Design–review.csv for current checks and open decisions.

Shared nets: two CAN backbones, power and ground only.
PRELIMINARY – incomplete board; not for fabrication
Team 4
Sheet: /
File: Lab4.kicad_sch
Title: Lab 4 – three–zone controller
Size: A2
Date: 2026–10–20
Rev: 0.3
KiCad E.D.A. 10.0.4
Id: 1/5

Front zone — CAN and servo current sensing

CAN A: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

CAN B: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

SERVO SHUNT — calculated proposal (not measured)

6.6 V regulator -> 10 mOhm shunt -> servo +
Servo GND returns directly to power ground.
Kelvin sense: separate thin traces from shunt pads to IN+/IN-.
Gain 50: Vout = 0.5 V/A. Proposed range 0-5 A -> 0-2.5 V.
12-bit / 3.3 V: 1.61 mA/count (ideal quantization).
At 5 A: shunt drop 50 mV; loss 0.25 W. Select >=1 W.
RC: 4.7k + 100n -> 339 Hz; proposed ADC 4 ksample/s.
Near-zero amplifier output clips; calibrate offset and gain.
This unidirectional circuit cannot measure regenerated current.
No servo idle/stall/noise measurements yet.
5 A measurement range is NOT a 5 A driver or regulator rating.

LAB3 header follows handout p17 top-to-bottom. Two GND pins first.
SRV/PWM_OUT probe actual actuator PWM; timing markers remain separate.
Physical header position / accessibility will be checked during PCB layout.

Servo and lamp circuits are below. Connector cable adapters still need physical verification.

STM32F401RET6 — independent HSI/PLL, no ST-LINK clock dependency

VCAP1 is the INTERNAL regulator output. Only its local capacitor connects here.
Keep every 100n beside its VDD pin; keep the VCAP loop short. DS10086 Fig.2 / Fig.18.

SWD: pin1 VTref senses target voltage; do not feed a second 3.3V supply.
UART header is TX/RX/GND, not RS-232. NRST pulls low to reset.
BOOT0 low boots flash; fit recovery shunt only for bootloader recovery.

TEST: active low on PC13, 10k/100n = 1ms release RC.
Firmware must debounce and meet the 10ms local-fault deadline.
Reset and self-test are separate buttons.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

SERVO AND VEHICLE LAMP OUTPUTS

Servo supply passes through R1. PWM uses PA9 and retains its reset pulldown.
J90 uses a new rated 3.96mm connector, not the existing servo plug.
Adapter must preserve GND / supply / signal; confirm the servo accepts 3.3V PWM.

3S LiPo BATTERY MONITOR: protected pack rail / 11 -> PA3 ADC1_IN3.
12.6V full -> 1.145V nominal ADC; 15V ceiling -> 1.364V.
Use calibrated VDDA, 480-cycle sample at <=21MHz; allow >=10ms filter settling.
No internal pullup/pulldown. Enable BOR; do not treat the MCU VBAT backup pin as pack input.
Firmware must inhibit drive at startup, stale/invalid samples and low battery.
This is a pack-voltage monitor, not per-cell protection or a hardware disconnect.

Engineering review; actuator/protection and mechanical sign-off pending.
PRELIMINARY – incomplete board; not for fabrication
Team 4
Sheet: /Front zone/
File: Front.kicad_sch
Title: Front zone – integrated logic and interfaces
Size: A0
Date: 2026-10-23
Rev: 0.3
KiCad E.D.A. 10.0.4
Id: 2/5

Rear zone — CAN and motor current sensing

CAN A: 16 MHz / 500 kbit/s / 3.3 V

LEFT MOTOR: L298 output -> ACS712 -> motor lead

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

LAB3 header follows handout p17 top-to-bottom. Two GND pins first.
SRV/PWM_OUT probe actual actuator PWM; timing markers remain separate.
Physical header position / accessibility will be checked during PCB layout.

CAN B: 16 MHz / 500 kbit/s / 3.3 V

RIGHT MOTOR: L298 output -> ACS712 -> motor lead

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

L298 and output circuits are below. Current limiting and motor operating envelope remains unresolved; heatsink geometry is specified; no motor-power sign-off.

Each motor: +/-5 A; VADC = 1.25 V + 0.0925 V/A * I.
Ideal 12-bit scale: 8.71 mA/count at VREF=3.3 V.
10k/10k divider keeps a 5 V sensor output at 2.5 V.
Filter source resistance: (10k || 10k) + 1k = 6k.
6k * 100n -> 265 Hz. Proposed ADC: 4 ksample/s/channel.
Stop motors before zero calibration; preserve signed readings.
Hall sensor has internal conductor, not an external shunt.
At 5 A and typical 1.2 mOhm, conductor loss = 30 mW.
Motor return lead connects to the OTHER L298 output, not GND.

STM32F401RET6 — independent HSI/PLL, no ST-LINK clock dependency

VCAP1 is the INTERNAL regulator output. Only its local capacitor connects here.
Keep every 100n beside its VDD pin; keep the VCAP loop short. DS10086 Fig.2 / Fig.18.

SWD: pin1 VTref senses target voltage; do not feed a second 3.3V supply.
UART header is TX/RX/GND, not RS-232. NRST pulls low to reset.
BOOT0 low boots flash; fit recovery shunt only for bootloader recovery.

TEST: active low on PC13, 10k/100n = 1ms release RC.
Firmware must debounce and meet the 10ms local-fault deadline.
Reset and self-test are separate buttons.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

ENCODER INPUTS: board pins 1=4.0V regulated, 2=GND, 3=A, 4=B. Verify actual cable order.
10k pullups support open-collector outputs; push-pull input must stay <=5.5V.
U81 at 3.3V gives Schmitt hysteresis and MCU-safe outputs; unused inputs grounded.
Both A and B invert. Check direction/counts after the PCB firmware port.

MOTOR POWER STAGE — cooling / stall behavior to verify

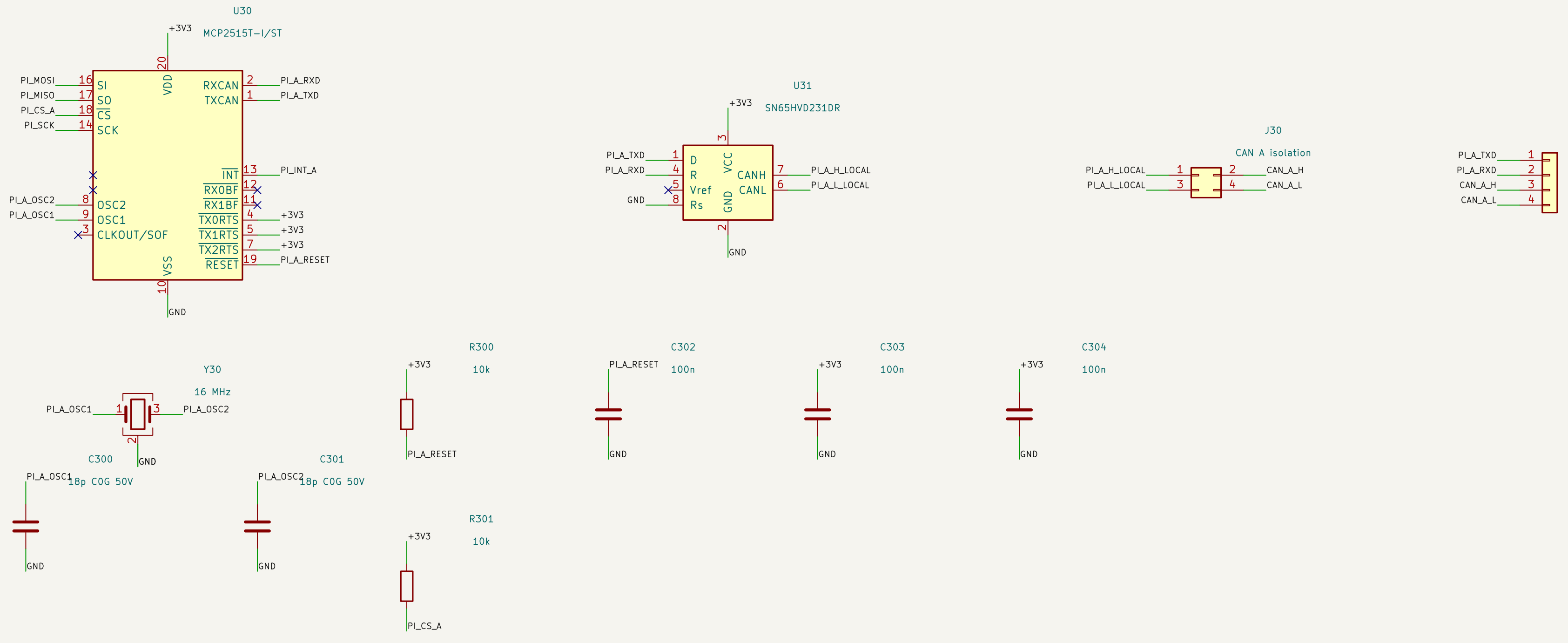
OUT1/OUT3 -> ACS712 IP+ -> IP- -> motor lead. Other lead -> OUT2/OUT4, NEVER ground.
ENA/ENB high + equal direction inputs gives dynamic braking. Enable low gives coast.
Existing 10k pulldowns keep enables low during reset. SENSE_A/B return to power ground.
This schematic has no autonomous motor-current limiter. A fuse and filtered ADC are not substitutes.
Keep the course L298N; check cooling and course stall-test expectations. 3.2A is a motor stall specification, not normal current.
The eight diodes clamp each bridge output to protected battery and ground. Keep loops short.

ENCODER SUPPLY: 5.1V -> U82 4.0V -> J82/J83 pin1.
40mA combined planning load, not a measurement.
Keep the Schmitt buffer and pullups at3.3V; signal inputs tolerate up to5.5V.
C804/C805 are polarized: positive pad1 goes to the named supply.

Engineering review; actuator/protection and mechanical sign-off pending.
PRELIMINARY — incomplete board; not for fabrication
Team 4
Sheet: /Rear zone/
File: Rear.kicad_sch
Title: Rear zone — integrated logic and interfaces
Size: A0
Date: 2026-10-23
Rev: 0.3
KiCad E.D.A. 10.0.4
Id: 3/5

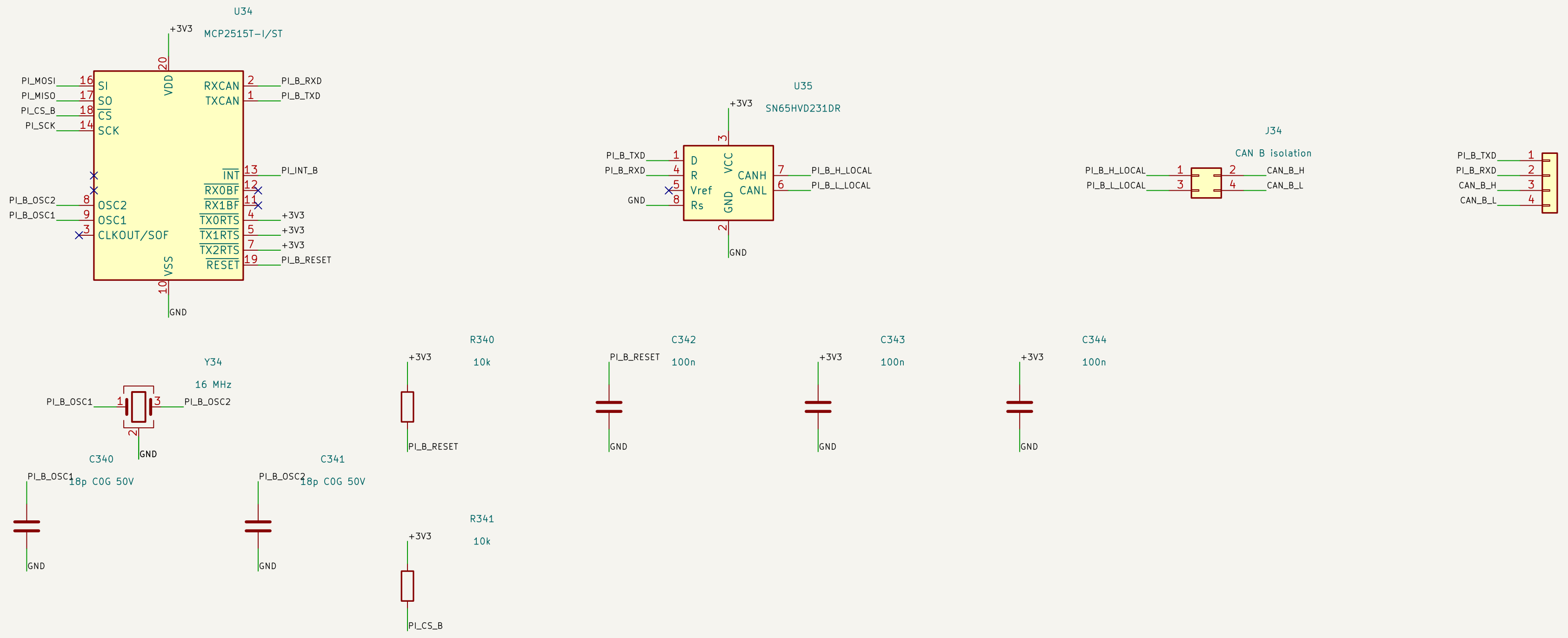
Cockpit zone — dual CAN

CAN A: 16 MHz / 500 kbit/s / 3.3 V



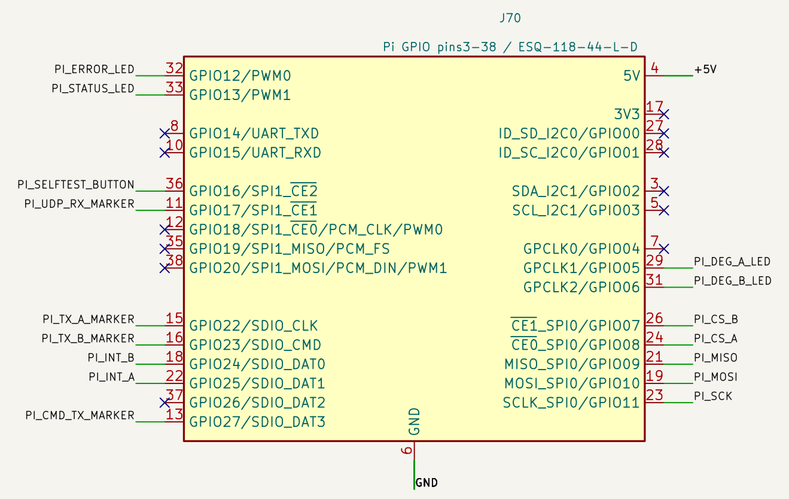
RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

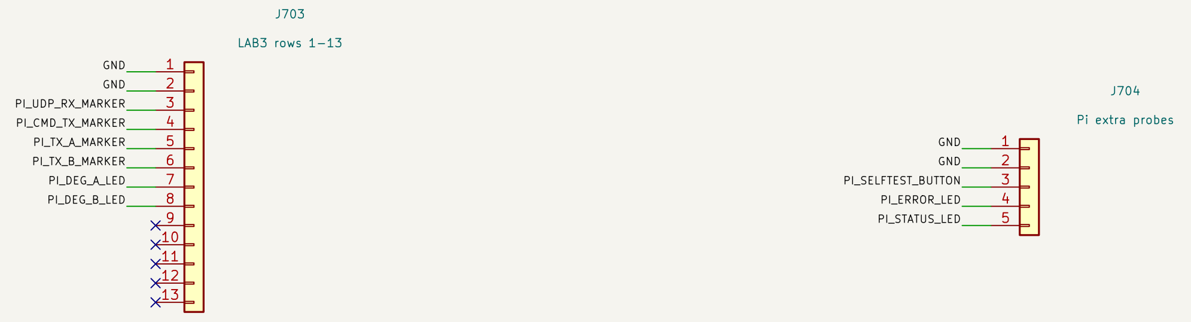
CAN B: 16 MHz / 500 kbit/s / 3.3 V



RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

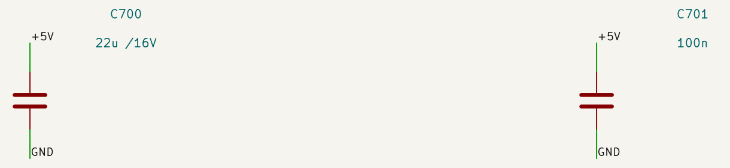
RASPBERRY PI 4 — socket covers physical pins3–38


J70
Pi GPIO pins3–38 / ESQ-118-44-L-D
PI_ERROR_LED 32 GPIO12/PWM0
PI_STATUS_LED 33 GPIO13/PWM1
8 GPIO14/UART_TXD
10 GPIO15/UART_RXD
PI_SELFTEST_BUTTON 36 GPIO16/SPI1_CE2
PI_UDP_RX_MARKER 11 GPIO17/SPI1_CE1
12 GPIO18/SPI1_CE0/PCM_CLK/PWM0
35 GPIO19/SPI1_MISO/PCM_FS
38 GPIO20/SPI1_MOSI/PCM_DIN/PWM1
PI_TX_A_MARKER 15 GPIO22/SDIO_CLK
PI_TX_B_MARKER 16 GPIO23/SDIO_CMD
PI_INT_B 18 GPIO24/SDIO_DAT0
PI_INT_A 22 GPIO25/SDIO_DAT1
37 GPIO26/SDIO_DAT2
PI_CMD_TX_MARKER 13 GPIO27/SDIO_DAT3
5V 4 +5V
3V3 17
ID_SD_I2C0/GPIO00 27
ID_SC_I2C0/GPIO01 28
SDA_I2C1/GPIO02 3
SCL_I2C1/GPIO03 5
GPCLK0/GPIO04 7
GPCLK1/GPIO05 29 PI_DEG_A_LED
GPCLK2/GPIO06 31 PI_DEG_B_LED
CE1_SPI0/GPIO07 26 PI_CS_B
CE0_SPI0/GPIO08 24 PI_CS_A
MISO_SPI0/GPIO09 21 PI_MISO
MOSI_SPI0/GPIO10 19 PI_MOSI
SCLK_SPI0/GPIO11 23 PI_SCK
GND 6
GND

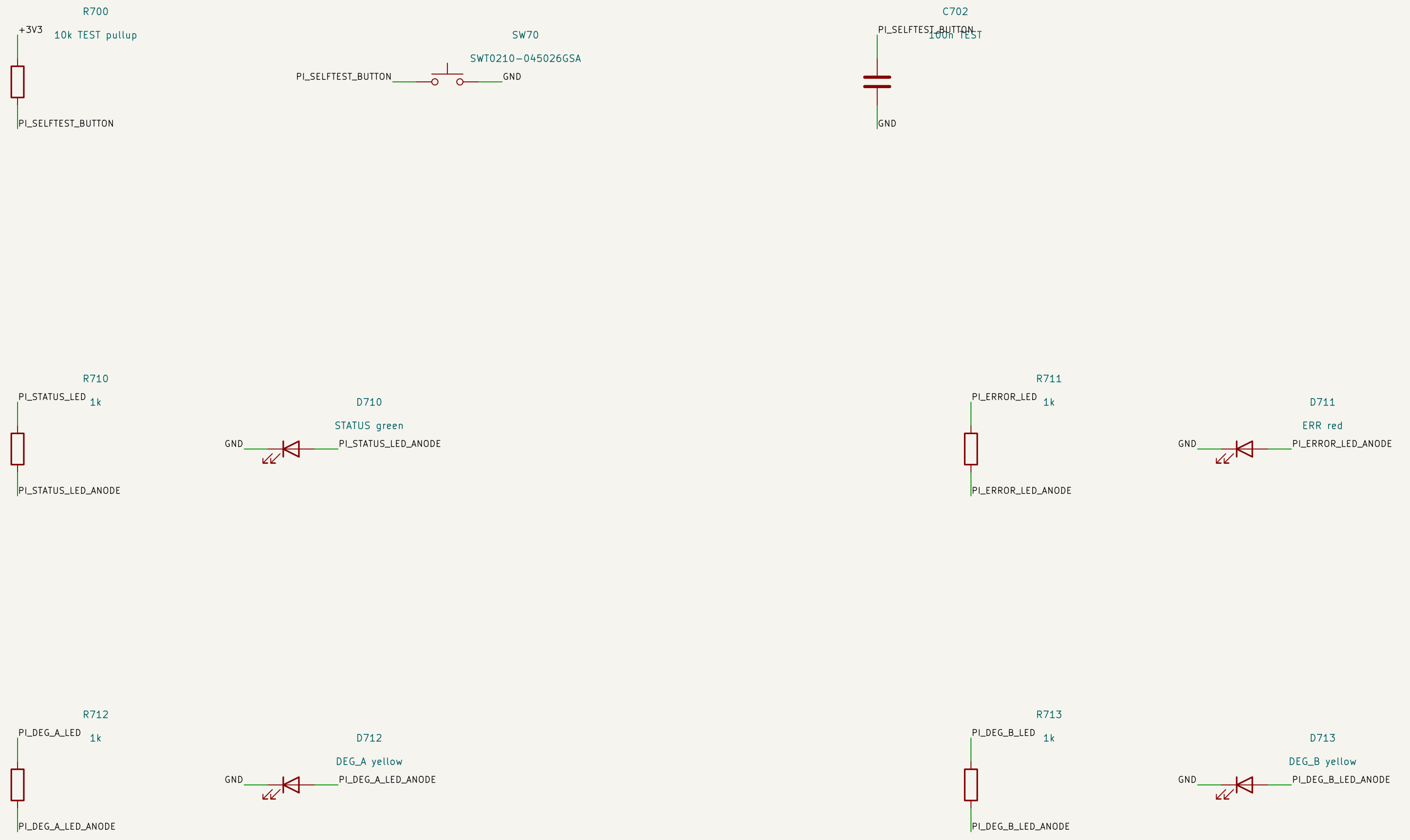


Rows 9–13 intentionally unused on Pi (Lab3 p17).
Two GND pins per probe group. Active–high LED outputs.
Physical reachability is a layout check, not established by this schematic.

Pi pin4 receives board5.1V. Socket omits1/2/39/40; Pi3.3V pin17 is NOT joined to board3.3V.
Do not attach independent USB–C power while board 5.1V is connected.
Align first socket pair to Pi3/4, not1/2. Check mating unpowered during assembly.

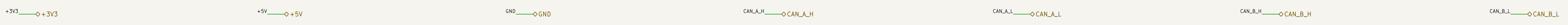


Local bypass is not a load–step proof. Pi bulk capacitance / rail transient test remains open.



All Pi GPIOs are 3.3V signals. Header pin number is NOT BCM GPIO number.
Independent board 3.3V supplies the six CAN interfaces.
A=CE0/IRQ25; B=CE1/IRQ24; confirm Linux labels against the installed overlay.

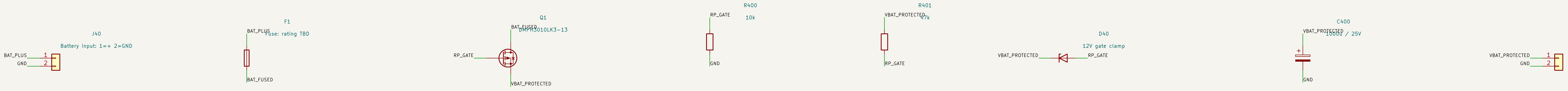
Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.



Engineering review; actuator/protection and mechanical sign–off pending.
PRELIMINARY — incomplete board; not for fabrication
Team 4
Sheet: /Cockpit zone/
File: Cockpit.kicad_sch
Title: Cockpit zone — integrated logic and interfaces
Size: A1 Date: 2026–10–23 **Rev: 0.3**
KiCad E.D.A. 10.0.4 Id: 4/5

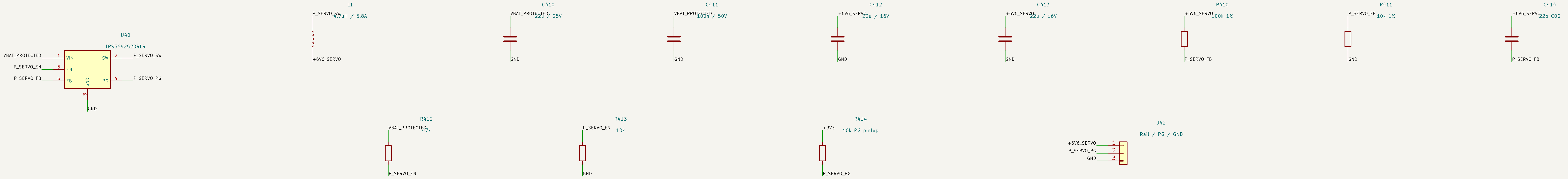
NiMH PACK INPUT — 10 cells; AA or Sub–C pack selection pending

Design input envelope: 8–15 V at the protected node; verify real cells, contact resistance and sag.
Battery / motor rail is UNREGULATED. No lithium charger or lithium BMS is included.



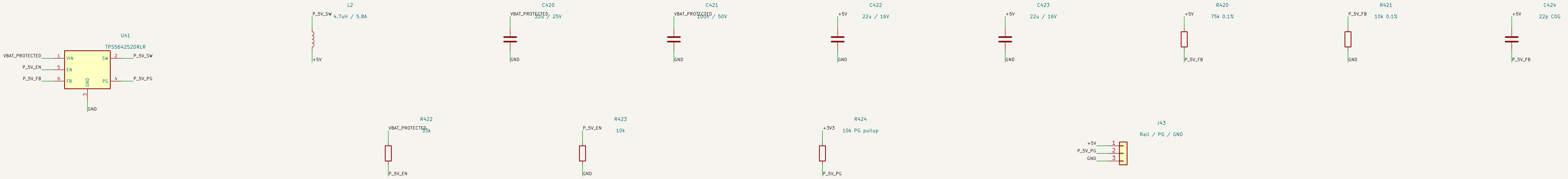
Q1 loss at 10 A: 1 W using 10 mOhm at 25 C; hotter silicon has higher resistance.
F1/holder/cable limits, transient overvoltage and thermal layout are open release checks.
J40 is Hirose DF63M–2P–3.96DSA; verify mating contacts, wire gauge, polarity and fuse coordination.

SERVO: 6.6 V; allocation 4 A; TPS564252 (4 A device rating)



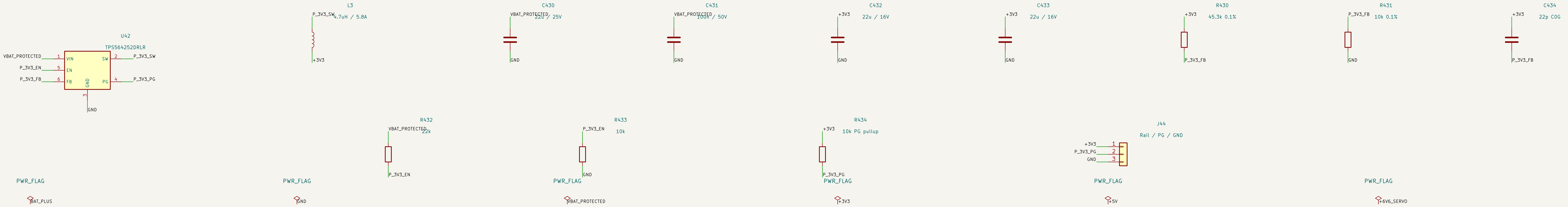
Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full–load thermal performance need validation.

5V: 5.1 V; allocation 3.3 A; TPS564252 (4 A device rating)

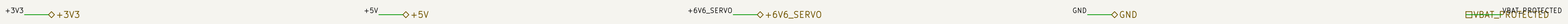


Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full–load thermal performance need validation.

3V3: 3.318 V; allocation 0.8 A; TPS564252 (4 A device rating)



Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full–load thermal performance need validation.



Engineering review; actuator/protection and mechanical sign–off pending.
PRELIMINARY — incomplete board; not for fabrication
Team 4

Sheet: /Power/
File: Power.kicad_sch

Title: Power — NiMH input and three buck converters

Size: A2	Date: 2026–10–23	**Rev: 0.3**
KiCad E.D.A. 10.0.4		Id: 5/5